Answers to 100 Short Questions

CLASS 11 • SUMMATIVE THEORY • 3 MARKS PER QUESTION

Numbering matches the companion SLO-based question bank.
For numericals, write the formula, substitution and final answer with its unit.

Measurements and mechanics

Q01–Q06

Q01 Base units of pressure

- **Force.** From F = ma, 1 N = 1 kg·m·s^{-2}.
- **Pressure.** Pressure = force ÷ area. Therefore 1 Pa = 1 N/m^2 = **1 kg·m^{-1}·s^{-2}**.

Q02 Checking dimensions

- **Compare the terms.** [ut] = (LT^{-1})(T) = L, but [½at] = (LT^{-2})(T) = LT^{-1}.
- **Correct the equation.** Use t^2 because [at^2] = (LT^{-2})(T^2) = L. The corrected equation is **s = ut + ½at^2**.

Q03 Accuracy and precision

- **More precise.** Balance B. Its readings have a smaller spread, 0.1 g compared with 0.2 g for A.
- **More accurate.** Balance A. Its mean is 50.0 g, equal to the reference value. B's mean is about 52.0 g.

Q04 Uncertainty in area

- **Area.** A = 20.0 × 10.0 = 200 cm^2.
- **Percentage uncertainty.** (0.2/20.0 + 0.1/10.0) × 100% = 2.0%.
- **Result.** ΔA = 0.020 × 200 = 4 cm^2, so **A = (200 ± 4) cm^2**.

Q05 Components of the pull

- **Horizontal.** F_x = 80 cos 30° = **69.3 N**.
- **Vertical.** F_y = 80 sin 30° = **40 N upward**. This upward component partly supports the trolley's weight, reducing the normal force.

Q06 Ball rolling off a ledge

- **Time to fall.** The initial vertical velocity is zero. From h = ½gt^2, t = √(2h/g) = √(40/10) = **2.0 s**.
- **Horizontal distance.** The horizontal speed stays constant, so x = 6.0 × 2.0 = **12 m**.

Rotational and circular motion

Q07–Q15

Q07 Angle and average angular speed

- **Angle.** $\theta = 150 \times \pi/180 = 5\pi/6$ rad $\approx$ **2.62 rad**.
- **Angular speed.** $\omega = \theta/t = (5\pi/6)/0.50 \approx$ **5.24 rad/s**.

Q08 Fan slowing down

- **Angular acceleration.** $\alpha = (\omega_f - \omega_i)/t = (8.0 - 24)/4.0 =$ **-4.0 rad/s²**.
- **Angular displacement.** $\theta = \frac{1}{2}(\omega_i + \omega_f)t = \frac{1}{2}(24 + 8.0)(4.0) =$ **64 rad**.

Q09 Two dots on a rotating disc

- **Linear speeds.** Use $v = r\omega$. The inner dot has $v = 0.040 \times 15 =$ **0.60 m/s**. The outer dot has $v = 0.12 \times 15 =$ **1.8 m/s**.
- **Angular speeds.** Both dots turn through the same angle in the same time because the disc is rigid.

Q10 Acceleration in circular motion

- **Why it accelerates.** Its speed is constant, but the direction of its velocity keeps changing. The acceleration is toward the centre of the circle.
- **After release.** It moves along the tangent at the release point, in the direction of its velocity at that instant.

Q11 Car on a flat bend

- **Source of force.** Static friction from the road on the tyres acts toward the centre of the bend.
- **Magnitude.** $F = mv^2/r = 900(12^2)/45 =$ **2,880 N**.

Q12 Apparent weightlessness in orbit

- **Observation.** The pen appears to float beside the astronaut just after release.
- **Reason.** Gravity accelerates the pen, astronaut and spacecraft together in orbital free fall. The astronaut therefore feels no supporting force from the floor.

Q13 Artificial gravity at the rim

- **Direction of down.** Down is radially outward, toward the floor at the rim.
- **Reason.** The floor pushes the crew inward, toward the axis, providing the centripetal force. The crew feels this support as apparent weight.

Q14 Moment of inertia and angular momentum

- **Moment of inertia.** $I = \Sigma mr^2 = 0.50(0.20^2) + 0.50(0.40^2) =$ **0.10 kg·m²**.
- **Angular momentum.** $L = I\omega = 0.10 \times 6.0 =$ **0.60 kg·m²/s**.

Q15 Net torque and angular acceleration

- **Net torque.** $\tau = 1.8 - 0.20 =$ **1.6 N·m** in the driving direction.
- **Angular acceleration.** $\alpha = \tau/I = 1.6/0.40 =$ **4.0 rad/s²**.

Rotational and circular motion

Q16 **Rotating stool**

- **Explanation.** Pulling the masses inward reduces I. With negligible external torque about the axis, Iω is conserved, so the angular speed increases.
- **Ratio.** $I_i\omega_i = I_f\omega_f$. Hence $\omega_f/\omega_i = I_i/I_f = 1/(3/4) =$ **4/3**.

Q17 **Separation in a centrifuge**

- **Collection point.** Denser particles collect at the outer end of the tube, farther from the axis.
- **Reason.** The inward pressure force from the rotating liquid is too small to provide the full centripetal force needed by a denser particle moving with the liquid. The particle therefore moves outward relative to the liquid.

Work and kinetic energy

Q18 **Deriving kinetic energy**

- **Work.** A constant net force along the displacement does work W = Fs = mas.
- **Substitute.** Starting from rest, $v^2 = 2as$, so $as = v^2/2$. Thus $W = \frac{1}{2}mv^2$. This work is the gain in kinetic energy, so **$K = \frac{1}{2}mv^2$**.

Q19 **Work from a force–displacement graph**

- **From 0 to 2 m.** Area of the first triangle = ½(2)(6) = **6 J**.
- **From 0 to 5 m.** Add the rectangle from 2 to 4 m and the triangle from 4 to 5 m. Total work = 6 + (2)(6) + ½(1)(6) = **21 J**.

Q20 **Work around a closed path**

- **Gravity.** Its net work is zero. Gravity does negative work on the way up and equal positive work on the way down.
- **Friction.** Its net work is negative because it opposes motion along the whole path. Its work depends on the distance travelled.

Q21 **Crate pulled across a rough floor**

- **Net work.** W = (20 − 8.0)(5.0) = **60 J**.
- **Final speed.** $W = \Delta K = \frac{1}{2}mv^2$ because the crate starts from rest. Hence $v = \sqrt{[2(60)/4.0]} \approx$ **5.5 m/s**.

Q22 **Cyclist slowing down**

- **Change in kinetic energy.** $\Delta K = \frac{1}{2}(60)(4.0^2 - 8.0^2) =$ **−1,440 J**.
- **Average resisting force.** On the level road, $-F(12) = \Delta K$. Thus F = 1,440/12 = **120 N**, opposite the motion.

Work and kinetic energy

Q23 Relative speed in an elastic collision

- **Relative speeds.** The approach speed is 5.0 + 2.0 = 7.0 m/s. In an elastic collision the separation speed is also 7.0 m/s.
- **Final speed.** If the second cart's speed is v, then 1.0 + v = 7.0, so it moves **6.0 m/s to the right**. Total kinetic energy is conserved.

Q24 Two trolleys sticking together

- **Final motion.** They come to rest together. Their initial momenta are equal and opposite, so the total momentum is zero before and after.
- **Energy.** Their initial kinetic energies are both positive. In the collision this kinetic energy is transferred to deformation, heating and sound.

Fluid mechanics

Q25 Density of a floating block

- **Force balance.** Upthrust equals weight. With block volume V, $\rho_{water} g(0.75V) = \rho_{block} gV$.
- **Density.** $\rho_{block} = 0.75\rho_{water} = 0.75 \times 1{,}000 =$ **750 kg/m³**.

Q26 Why a steel ship floats

- **Hollow hull.** The hull encloses air, making the ship's average density lower than seawater's. It can displace enough water for upthrust to balance its weight. Solid steel is denser than seawater and sinks.
- **Added cargo.** The ship sits deeper and displaces a larger volume of water to balance the increased weight.

Q27 Assumptions about an ideal fluid

- **Incompressible.** Its density stays constant as pressure changes.
- **Non-viscous.** There is no internal friction between neighbouring fluid layers. Real oil layers exert drag, so the non-viscous assumption fails.

Q28 Flow rate and speed in a pipe

- **Flow rate.** $v_1 = 0.60$ m/s = 60 cm/s. $Q = A_1v_1 = 12 \times 60 =$ **720 cm³/s**.
- **Speed in the narrower section.** Q is constant, so $v_2 = Q/A_2 = 720/4.0 =$ **180 cm/s (1.8 m/s)**.

Q29 Hose outlet speed

- **Prediction.** The water leaves at three times its original speed.
- **Reason.** The same mass of water leaves per second, and its density is constant. Therefore $A_1v_1 = A_2v_2$. With $A_2 = A_1/3$, $\mathbf{v_2 = 3v_1}$.

Fluid mechanics

Q30 Pressure drop in a vertical pipe

- **Speed.** For steady incompressible flow, Av is constant. Equal cross-sectional areas give equal speeds.
- **Pressure drop.** The kinetic terms cancel in Bernoulli's equation. $P_{lower} - P_{upper} = \rho g \Delta h = 1{,}000 \times 10 \times 2.0 =$ **20,000 Pa**.

Q31 Viscous drag on a bead

- **Greater drag.** At the same bead speed, glycerine exerts greater viscous drag because its viscosity is higher.
- **Reason.** The bead pulls nearby liquid layers along. These layers move relative to more distant layers, and internal friction opposes their relative motion. This produces a force on the bead opposite its motion.

Q32 Vortices and superfluidity

- **Ordinary liquid.** Viscous friction between layers transfers the energy of the vortex into internal energy, so the motion dies away.
- **Idealized superfluid.** Its viscosity is zero. A vortex can persist because there is no viscous friction to slow it.

Q33 Venturi filter pump

- **At the throat.** The smaller area makes the water flow faster, so its static pressure is lower at the throat.
- **Air intake.** If throat pressure is below the air pressure at the inlet, the pressure difference drives air through the side tube into the water stream.

Deformation of solids

Q34 Structures of solids

- **First sample.** A crystalline solid has a repeating three-dimensional arrangement.
- **Second sample.** Long molecular chains are characteristic of a polymeric solid.
- **Glass.** Ordinary glass is amorphous and lacks the long-range repeating order of a crystal.

Q35 Compressive strain

- **Type.** The rod undergoes compressive deformation because the forces shorten it.
- **Strain.** Its magnitude is $|\Delta L|/L = (0.20 \times 10^{-3})/0.50 =$ **4.0×10^{-4}**. Strain has no unit because it is a ratio of lengths.

Deformation of solids

Q36 Stress, strain and Young modulus

- **Stress.** F/A = 50/(1.0 × 10^{-6}) = **5.0 × 10^{7} Pa**.
- **Strain.** ΔL/L = (1.0 × 10^{-3})/2.0 = **5.0 × 10^{-4}** (no unit).
- **Young modulus.** Y = stress/strain = (5.0 × 10^{7})/(5.0 × 10^{-4}) = **1.0 × 10^{11} Pa**.

Q37 Extension of two wires

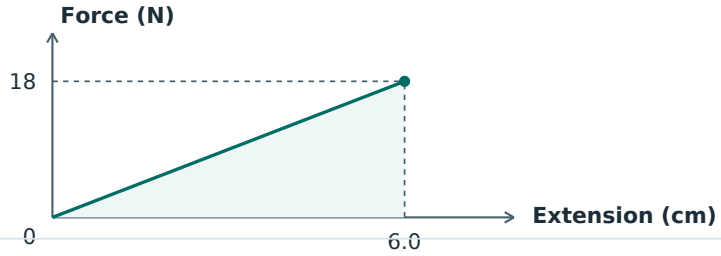
- **Strain.** Both wires have the same stress F/A and the same Young modulus. Since strain = stress/Y, their strains are equal.
- **Extension.** ΔL = strain × original length. Wire A is twice as long, so its extension is **twice that of wire B**.

Q38 Recovered and permanent extension

- **Recovered extension.** 3.0 − 0.80 = **2.2 mm**.
- **Elastic limit.** The remaining 0.80 mm is permanent (plastic) deformation. This shows that the wire was stretched beyond its elastic limit.

Q39 Energy from a force–extension graph

- **Sketch.** Draw a straight line from the origin to (6.0 cm, 18 N) because force is proportional to extension.
- **Stored energy.** It is the triangular area under the graph. E = ½Fx = ½(18)(0.060) = **0.54 J**.



Q40 Effect of extension on elastic energy

- **Relationship.** For the same spring, E = ½kx^2, so energy is proportional to x^2.
- **Calculation.** The extension triples. Therefore E_2 = 0.20 × (6.0/2.0)^2 = **1.8 J**.

Heat and thermodynamics

Q41–Q46

Q41 Thermal equilibrium and internal energy

- **On contact.** There is no net heat transfer because both blocks have the same temperature. They are in thermal equilibrium even though their masses differ.
- **On heating.** The block's internal energy increases. Its particles gain energy and vibrate more vigorously as its temperature rises.

Q42 Two forms of the ideal gas equation

- **Pressure.** $p = nRT/V = (0.50)(8.31)(300)/0.020 \approx$ **6.2×10^4 Pa**.
- **Molecular form.** $pV = NkT$, where N is the number of molecules. The Boltzmann constant is **$k = R/N_A$**, where N_A is Avogadro's constant.

Q43 When a real gas behaves ideally

- **Conditions.** High temperature and low pressure.
- **Low pressure.** The molecules are far apart, so their own volume is very small compared with the volume of the gas.
- **High temperature.** The molecules have high average kinetic energy, so intermolecular attractions have little effect on their motion.

Q44 Expansion at constant pressure

- **Work done by the gas.** $W = p\Delta V = (1.0 \times 10^5)(0.014 - 0.010) =$ **400 J**.
- **Internal energy.** From $Q = \Delta U + W$, $\Delta U = Q - W = 550 - 400 =$ **+150 J**.

Q45 Compression with heat loss

- **Signs.** $Q = -120$ J because heat leaves the gas. $W = -200$ J because the piston does work on the gas, and W denotes work done by the gas.
- **Internal energy.** $\Delta U = Q - W = -120 - (-200) =$ **+80 J**.

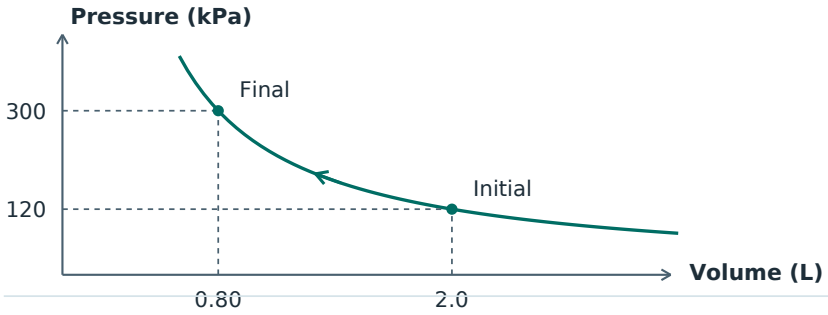
Q46 Heating a gas at constant volume

- **Prediction.** The gas pressure increases.
- **Reason.** The molecules move faster at the higher temperature. They hit the walls more often and transfer more momentum per collision. The force per unit area therefore increases.

Heat and thermodynamics

Q47 Isothermal compression

- **Final pressure.** At constant temperature, $p_1V_1 = p_2V_2$. Thus $p_2 = (120 \times 2.0)/0.80 =$ **300 kPa**.
- **Graph.** The curve is a downward-sloping hyperbola because $p \propto 1/V$. Compression takes the gas from (2.0 L, 120 kPa) to (0.80 L, 300 kPa).



Q48 Irreversible heat transfer

- **Direction.** Heat flows from the hot block to the cooler water until they reach the same temperature.
- **Second law.** Heat flows spontaneously from higher to lower temperature. The total entropy increases. The transfer cannot reverse by itself and restore the original temperatures.

Q49 Heat rejected by an engine

- **Heat rejected.** The internal energy returns to its starting value after each cycle. Thus $Q_{cold} = Q_{hot} - W = 800 - 300 =$ **500 J**.
- **Second law.** A cyclic engine must reject some heat to a colder reservoir. It cannot take heat from only one reservoir and convert it all into work.

Q50 Carnot efficiency limit

- **Maximum efficiency.** $\eta = 1 - T_{cold}/T_{hot} = 1 - 300/600 = 0.50 =$ **50%**.
- **Decision.** The claim of 60% is impossible because it exceeds the Carnot limit for those reservoir temperatures.

Q51 Energy degradation by friction

- **Energy transfer.** Bearing friction changes the flywheel's rotational kinetic energy into internal energy of the bearings and their surroundings.
- **Degradation.** The energy spreads as heat, increasing total entropy. The dispersed thermal energy is less available to do useful work than the original rotational energy.

Waves

Q52–Q58

Q52 Wave intensity and amplitude

- **Initial intensity.** $I = P/A = 6.0/0.50 =$ **12 W/m²**.
- **New intensity.** For the same conditions, $I \propto \text{amplitude}^2$. Halving the amplitude gives $I' = 12(½)^2 =$ **3.0 W/m²**.

Q53 Doppler shift of a receding siren

- **Prediction.** The observed frequency is lower because the source is moving away and the wavefronts reaching the listener are farther apart.
- **Calculation.** For a receding source, $f_o = f_s v/(v + v_s) = (680)(340)/(340 + 20) \approx$ **642 Hz**.

Q54 Doppler shift in weather radar

- **Frequency.** The returning echo has a higher frequency because the raindrops are moving toward the radar.
- **Speed.** The radar compares the echo frequency with the transmitted frequency. For the same radar frequency, faster motion along the beam produces a larger shift.

Q55 Polarization of transverse waves

- **Phenomenon.** Polarization. The vertical slit selects vertical oscillations of the transverse rope wave.
- **Sound in air.** In a longitudinal sound wave the particles vibrate parallel to the direction of travel, so there is no transverse direction of vibration for a polarizer to select.

Q56 Two analysers and Malus's law

- **First analyser.** $I_1 = I_0 \cos^2 30° = 80(\sqrt{3}/2)^2 =$ **60 W/m²**. The emerging light is polarized along the first axis.
- **Second analyser.** Its axis is 90° to that polarization, so $I_2 = 60 \cos^2 90° =$ **0 W/m²** for ideal analysers.

Q57 Interference at X and Y

- **At X.** The path difference is 1½λ, an odd multiple of λ/2. The waves arrive in opposite phase and cancel because their amplitudes are equal. This is destructive interference.
- **At Y.** The path difference is 2λ, a whole number of wavelengths. The waves arrive in phase and reinforce. This is constructive interference.

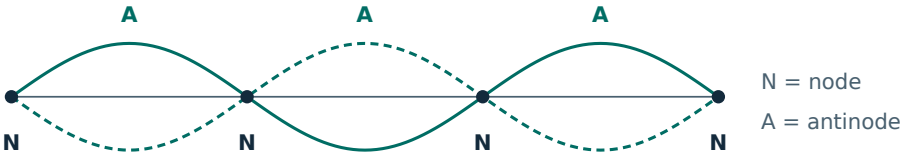
Q58 Noise-cancelling headphones

- **Phase difference.** The added wave should differ in phase by **180° (π rad)** at the ear.
- **Explanation.** A crest of one wave meets a trough of the other. By superposition, equal and opposite displacements add to zero, so the hum is reduced.

Waves

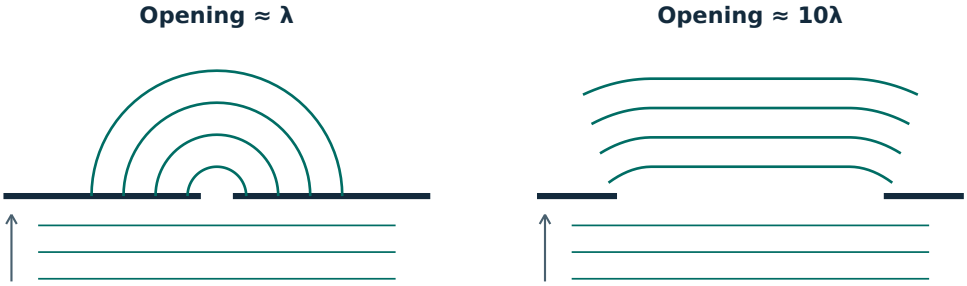
Q59 Third harmonic on a string

- **Sketch.** Draw three loops between the fixed ends. There are four nodes, including both ends, and three antinodes, one at the centre of each loop.
- **Wavelength.** Each loop is half a wavelength. Hence L = 3λ/2, so **λ = 2L/3**.



Q60 Diffraction at narrow and wide gaps

- **Gap ≈ λ.** Diffraction is strong. The emerging wavefronts are nearly semicircular and spread widely.
- **Gap ≈ 10λ.** Diffraction is weak. The middle parts of the emerging wavefronts are nearly straight, with slight bending near the edges.



Wavefronts approach the barriers from below.

Q61 Coherent sources

- **Condition.** Coherent waves have the same frequency and maintain a constant phase difference.
- **Why split one source?** The phase difference between independent lamps varies randomly, making the pattern unstable. Two beams split from one source share its phase changes, so fixed paths can maintain a constant phase difference.

Q62 Using beats to find a string frequency

- **Possible frequencies.** The beat frequency is $|f_{string} - 440| = 3$ Hz, so the possibilities are **437 Hz or 443 Hz**.
- **Decision.** Raising 443 Hz to 444 Hz increases the beat rate to 4 Hz. Raising 437 Hz to 438 Hz would lower it to 2 Hz. The original frequency was **443 Hz**.

Electrostatics

Q63–Q67

Q63 Field from the force on a test charge

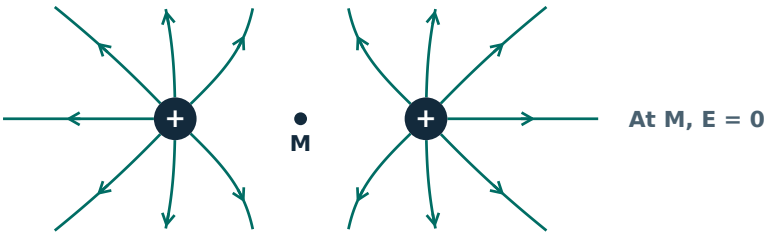
- **Magnitude and direction.** $E = F/q = 0.012/(3.0 \times 10^{-6}) =$ **4.0×10^{3} N/C east**. The field points in the direction of the force on a positive test charge.
- **Negative charge.** Its force is opposite to the field, so it acts **west**.

Q64 Electric field between plates

- **Magnitude.** $E = \Delta V/d = 24/(4.0 \times 10^{-3}) =$ **6.0×10^{3} V/m**.
- **Electron's motion.** The field points from the positive plate to the negative plate. An electron has negative charge, so its force and acceleration are toward the **positive plate**.

Q65 Field of two equal positive charges

- **Sketch.** The field lines point outward from both charges and curve away from the region between them. No field line joins the two positive charges.
- **Midpoint.** It is equally far from each charge. The two fields there have equal magnitudes and opposite directions, so the net field is zero.



Q66 Field outside a conducting sphere

- **Model.** At this exterior point the charged sphere acts as a point charge at its centre, 0.30 m away.
- **Calculation.** $E = kQ/r^2 = (9.0 \times 10^{9})(4.0 \times 10^{-9})/(0.30)^2 =$ **4.0×10^{2} N/C**. The field points away from the positive sphere.

Q67 Electrostatic shielding

- **Charge redistribution.** The charged rod moves free electrons in the metal. Opposite charge is induced nearer the rod and like charge farther away, on the outer surface.
- **Inside the cavity.** At electrostatic equilibrium the closed conductor shields the empty cavity. The induced charges cancel the effect of the external field there, so **$E = 0$**.

Electrostatics

Q68 **Change in Coulomb force**

- **Magnitude.** By Coulomb's law, $F \propto |q_1 q_2|/r^2$. Tripling one charge and doubling the distance multiplies the force by $3/2^2$. $F' = (3/4)(0.18) =$ **0.135 N** (about 0.14 N).
- **Nature of force.** It remains repulsive because the signs of the charges are unchanged.

Q69 **Field halfway between opposite charges**

- **Direction.** At the midpoint, the field from the positive charge points away from it, and the field from the negative charge points toward it. Both point **toward the negative charge**.
- **Magnitude.** Each charge is 0.10 m away. $E_{net} = 2k|q|/r^2 = 2(9.0 \times 10^9)(2.0 \times 10^{-9})/(0.10)^2 =$ **3.6×10^3 N/C**.

Current electricity

Q70 **Drift speed and wire area**

- **Relation.** $I = nqAv_d$, where n is the carrier number density and q is the magnitude of the charge per carrier.
- **Ratio.** I, n and q are fixed, so $v_d \propto 1/A$. Doubling A makes $v'_d/v_d =$ **½**.

Q71 **Cell e.m.f. and lamp p.d.**

- **Cell e.m.f.** Energy supplied per coulomb = 9.0/3.0 = **3.0 V**.
- **Lamp p.d.** Energy received per coulomb = 6.0/3.0 = **2.0 V**.
- **Difference.** The remaining 1.0 J per coulomb is transferred elsewhere in the circuit, for example in the cell's internal resistance.

Q72 **Power of two heaters**

- **Power.** For the same voltage, $P = V^2/R$. The 4.0 Ω heater uses $12^2/4.0 =$ **36 W**. The 8.0 Ω heater uses $12^2/8.0 =$ **18 W**.
- **Energy in one minute.** $E = Pt$. The 4.0 Ω heater transfers 2,160 J, twice the 1,080 J transferred by the 8.0 Ω heater.

Q73 **Resistance of a stretched wire**

- **Area.** Volume = AL stays constant. When the length doubles, the cross-sectional area becomes **A/2**.
- **Resistance.** $R' = \rho(2L)/(A/2) = 4\rho L/A = 4R =$ **20 Ω**.

Q74 **Terminal potential difference**

- **With current.** $V = \varepsilon - Ir = 6.0 - (2.0)(0.50) =$ **5.0 V**.
- **Open circuit.** The current becomes zero, so there is no internal voltage drop. The terminal p.d. rises to the e.m.f., **6.0 V**.

Current electricity

Q75 **Kirchhoff’s rules and conservation**

- **Junction current.** Total current entering equals total current leaving. The other branch carries 4.0 − 1.5 = **2.5 A away from the junction**.
- **Conservation laws.** The junction rule follows from conservation of charge. The loop rule follows from conservation of energy.

Q76 **Branch currents in a parallel circuit**

- **Loop rule.** Each branch has the full 9.0 V across it. Thus $9.0 - 6.0I_1 = 0$ and $9.0 - 3.0I_2 = 0$, giving $\mathbf{I_1 = 1.5\ A}$ and $\mathbf{I_2 = 3.0\ A}$.
- **Junction rule.** $I_{total} = I_1 + I_2 = \mathbf{4.5\ A}$.

Q77 **Potentiometer balance**

- **Principle and calculation.** The potential drop per unit length is constant, so the balancing e.m.f. is proportional to the balance length. $\varepsilon_2 = 1.50(48/60) = \mathbf{1.20\ V}$.
- **Null reading.** The galvanometer shows zero current because the potential drop along the balancing length equals the test cell’s e.m.f.

Q78 **LDR in a potential divider**

- **Prediction.** The output voltage across the LDR decreases as the light gets brighter.
- **Reason.** Brighter light lowers the LDR resistance. In series, the fraction of the supply across the LDR is $R_{LDR}/(R_{fixed} + R_{LDR})$, which falls when R_{LDR} falls.

Magnetism and electromagnetic induction

Q79 **Magnetic force on a wire**

- **Magnitude.** The wire is at right angles to the field. $F = BIL \sin 90° = (0.30)(5.0)(0.40) = \mathbf{0.60\ N}$.
- **Direction.** With the first finger pointing north (field) and the second finger east (current), the left thumb points **vertically upward**.

Q80 **Magnetic forces on a proton and an electron**

- **Magnitude.** The forces are equal because $F = |q|vB$ at 90° and both particles have the same $|q|$ and speed in the same field.
- **Direction.** The proton is positive and the electron negative. Reversing the sign of the charge reverses the magnetic force for the same velocity, so they initially bend in opposite directions.

Q81 **Motion in a uniform magnetic field**

- **Path.** The particle follows a circle because the magnetic force is perpendicular to its velocity and provides the centripetal force.
- **Speed.** A perpendicular force does no work, so kinetic energy and speed stay constant. The direction of velocity changes continuously.

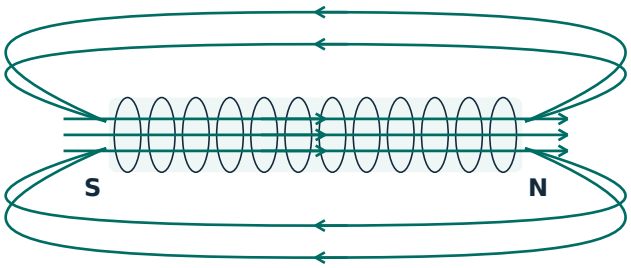
Magnetism and electromagnetic induction

Q82 Undeflected speed in a velocity selector

- **Balance.** The electric and magnetic forces cancel when $|q|E = |q|vB$. Therefore $v = E/B = (1.2 \times 10^4)/0.040 = \mathbf{3.0 \times 10^5\ m/s}$.
- **Faster particle.** The magnetic force is larger because $|q|vB$ increases with speed while $|q|E$ stays fixed.

Q83 Magnetic field of a solenoid

- **Sketch.** Inside, draw closely spaced, nearly parallel lines from S toward N. Outside, draw more widely spaced lines curving back from N to S.
- **Iron core.** Soft iron becomes magnetized and increases the field strength inside the solenoid at the same current.



Q84 Parallel wires with opposite currents

- **Result.** The wires repel each other.
- **Reason.** Use the right-hand grip rule to find the circular field of the first wire at the second wire. With the second wire's current in the opposite direction, Fleming's left-hand rule gives a force away from the first wire. The first wire is pushed away in the opposite direction.

Q85 Magnetic flux and flux linkage

- **Flux through one turn.** The field is perpendicular to the coil's plane, so $\Phi = BA = (0.15)(0.020) = \mathbf{3.0 \times 10^{-3}\ Wb}$.
- **Flux linkage.** $N\Phi = 50(3.0 \times 10^{-3}) = \mathbf{0.15\ Wb{\cdot}turn}$.

Q86 Induction when the flux decreases

- **Induced e.m.f.** From Faraday's law, $|\varepsilon| = N|\Delta\Phi|/\Delta t = 100(0.0060 - 0.0020)/0.20 = \mathbf{2.0\ V}$.
- **Direction.** In a closed coil the induced current produces a field in the same direction as the original field. By Lenz's law it opposes the *decrease* in flux.

Q87 Magnet moving toward a coil

- **Approaching magnet.** The near face becomes a north pole. It repels the approaching north pole, opposing the increase in flux through the coil.
- **Stationary magnet.** Once it is held still, the flux stops changing. There is no induced e.m.f., and the induced current falls to zero.

Special relativity

Q88–Q92

Q88 Inertial and non-inertial bus frames

- **Inertial phase.** The cabin is approximately an inertial frame while the bus travels straight at constant speed.
- **Braking.** The cabin decelerates, so it is then a non-inertial frame. The loose bag tends to keep its original forward velocity by inertia, so it moves forward relative to the slowing bus.

Q89 Speed of a light pulse

- **Measurements.** The spacecraft observer and the Earth observer both measure the speed of the pulse as **$c = 3.0 \times 10^8$ m/s** in vacuum.
- **Postulate.** All inertial observers measure the same speed of light in vacuum, regardless of the motion of the light source or observer.

Q90 Relativity of simultaneity

- **Conclusion.** The flashes are not simultaneous in the moving train frame because they occur at different places along the direction of motion.
- **Reason.** In the platform frame, $\Delta t = 0$ but the separation Δx is nonzero. The relativity formula gives $\Delta t' = \gamma(\Delta t - v\Delta x/c^2) = -\gamma v \Delta x/c^2$, which is nonzero. Thus, for a train moving to the right, the right-end flash occurs first in the train frame.

Q91 Time dilation for a moving clock

- **Proper time.** Δt_0 = **6.0 µs** because the two emissions occur at the same place in the probe frame.
- **Earth-frame interval.** $\Delta t = \Delta t_0/\sqrt{[1 - (v/c)^2]} = 6.0/\sqrt{(1 - 0.80^2)}$ = **10 µs**.

Q92 Length contraction of a probe

- **Proper length.** L_0 = **40 m** because it is measured in the probe's rest frame.
- **Laboratory length.** $L = L_0\sqrt{[1 - (v/c)^2]} = 40\sqrt{(1 - 0.60^2)}$ = **32 m**.

Nuclear and particle physics

Q93–Q94

Q93 Nucleon and proton numbers after alpha decay

- **Daughter nucleus.** The nucleon number is $226 - 4$ = **222**. The proton number is $88 - 2$ = **86**. Both nucleon number and charge are conserved.
- **Alpha particle.** It is a helium nucleus made of two protons and two neutrons.

Q94 Continuous beta-minus spectrum

- **Other particle.** An electron antineutrino is emitted along with the electron.
- **Energy sharing.** The electron and antineutrino can take different shares of the available energy in different decays, with a small share going to nuclear recoil. The electron therefore has a continuous range of possible energies.

Nuclear and particle physics

Q95 **Quark charges of the proton and neutron**

- **Proton (uud).** $(+\frac{2}{3} + \frac{2}{3} - \frac{1}{3})e = \mathbf{+e}$.
- **Neutron (udd).** $(+\frac{2}{3} - \frac{1}{3} - \frac{1}{3})e = \mathbf{0}$. Its quarks are charged, but their charges add to zero.

Q96 **Baryon, meson and lepton**

- **Neutron.** A baryon because it is made of three quarks (udd).
- **Bound up–antidown pair.** A meson because it consists of one quark and one antiquark.
- **Electron neutrino.** A lepton because it is a fundamental neutrino, with no quark constituents.

Q97 **Quark change in beta-plus decay**

- **Quark change.** One up quark changes to a down quark, so the proton (uud) becomes a neutron (udd).
- **Charged particle.** A positron (e^+) is emitted.
- **Neutral lepton.** An electron neutrino (ν_e) is emitted.

Q98 **Exchange particles**

- **Electromagnetic interaction.** The photon mediates repulsion between electrons.
- **Strong interaction between quarks.** A gluon carries the interaction.
- **Weak interaction in beta decay.** A charged W boson carries the interaction (W^- in β^- decay or W^+ in β^+ decay).

Q99 **Higgs field and elementary particle mass**

- **Associated particle.** The Higgs boson is a quantum of the Higgs field.
- **Mass.** An elementary particle such as an electron interacts with the Higgs field. This interaction gives the electron its rest mass.

Q100 **Electron–positron annihilation**

- **Comparison.** The electron and positron have the same mass and opposite charges ($-e$ and $+e$).
- **Products and energy.** They can produce two gamma-ray photons travelling in opposite directions in their centre-of-momentum frame. Their combined energy equals the initial rest energy, $\mathbf{2m_ec^2}$, when the initial kinetic energy is negligible.